JURNAL PRAKTIKUM

Antena dan Propogasi Gelombang

Kelompok : 3

Anggota :
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VSWR
(Voltage Standing Wave Ratio)

$$Loss = Att \times \ell$$
$$= 0{,}07 \times 50 = 3{,}5\ dB$$
$$\approx 2{,}24\ Watt$$

※ Frekuensi 140 MHz

$P_{in} = 9$ Watt

o> Ujung Kabel

$$VSWR = 1{,}3$$
$$\Gamma = \frac{VSWR - 1}{VSWR + 1} = \frac{0{,}3}{2{,}3} = 0{,}130$$
$$\Gamma^2 = \frac{P_{ref}}{P_{in}}$$
$$P_{ref} = P_{in} \times \Gamma^2$$
$$= 9 \times (0{,}130)^2 = 0{,}1521$$

o> Ujung Antena

$$P_{in\,(2)} = \frac{P_{in}}{Loss} = \frac{9}{2{,}24} = 4{,}01\ W$$
$$P_{ref\,(2)} = P_{ref} \times Loss$$
$$= 0{,}1521 \times 2{,}24 = 0{,}340\ W$$
$$\Gamma = \sqrt{\frac{P_{ref}}{P_{in}}} = \sqrt{\frac{0{,}340}{4{,}01}} = 0{,}289$$
$$VSWR_{(2)} = \frac{1 + \Gamma}{1 - \Gamma} = \frac{1 + 0{,}289}{1 - 0{,}289} = 1{,}812$$
$$Error = \frac{2{,}7 - 1{,}812}{2{,}7} = 0{,}328$$
$$= 32{,}8\ \%$$

* Frekuensi 147,5 MHz

P_{in} : 7,5 Watt

o> Ujung Kabel

VSWR : 1,1

$\Gamma = \frac{VSWR - 1}{VSWR + 1} = \frac{0,1}{2,1} = 0,0476$

$P_{ref} = P_{in} \times \Gamma^2$

$= 7,5 \times (0,0476)^2 = 0,0169$

o> Ujung Antena

$P_{in(2)} = \frac{P_{in}}{Loss} = \frac{7,5}{2,24} = 3,348$

$P_{ref(2)} = P_{ref} \times Loss$

$= 0,0169 \times 2,24 = 0,378$

$\Gamma = \sqrt{\frac{P_{ref}}{P_{in}}} = \sqrt{\frac{0,378}{3,348}} = 0,336$

$VSWR = \frac{1 + \Gamma}{1 - \Gamma} = \frac{1 + 0,336}{1 - 0,336} = 2,012.$

$Error = \frac{4 - 2,012}{4} * 100\% = 49,7\ \%$

* Frekuensi 136,250 MHz

P_{in} : 8,2 Watt

o> Ujung Kabel

VSWR : 1,4

$\Gamma = \frac{VSWR - 1}{VSWR + 1} = \frac{0,4}{2,4} = 0,167$

$P_{ref} = P_{in} \times \Gamma^2$

$= 8,2 \times (0,167)^2 = 0,229$

o> Ujung Antena

$P_{in(2)} = \frac{P_{in}}{Loss} = \frac{8,2}{2,24} = 3,66$

$P_{ref(2)} = P_{ref} \times Loss$

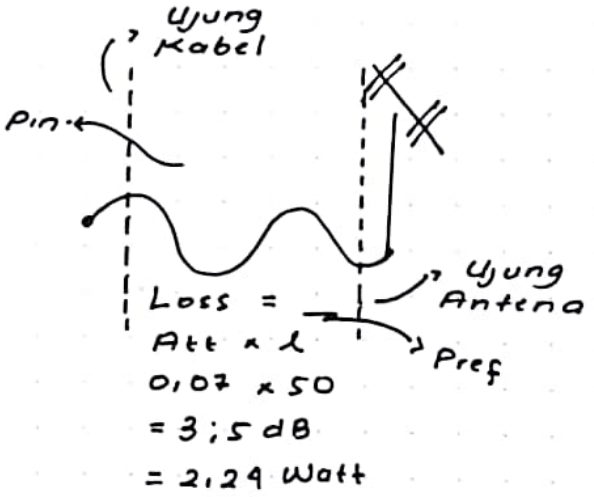
$= 0,229 \times 2,24 = 0,513$

$\Gamma = \sqrt{\frac{P_{ref}}{P_{in}}} = \sqrt{\frac{0,513}{3,66}} = 0,374$

$VSWR = \frac{1 + \Gamma}{1 - \Gamma} = \frac{1 + 0,374}{1 - 0,374} = 2,195$

$Error = \frac{2,195 - 1,7}{2,195} * 100\% = 22,55\ \%$

Blok Diagram



Tabel

* Ujung Kabel

Parameter	140 MHz	147,5 MHz	136,25 MHz
Pin	9 Watt	7,5 Watt	8,2 Watt
Pref	0,1521 Watt	0,0169 Watt	0,229 Watt
Γ	0,130	0,0476	0,167
VSWR	1,3	1,1	1,4

* Ujung Antena

Parameter	140 MHz	147,5 MHz	136,25 MHz
Pin	9,01 W	3,348 W	3,66 W
Pref	0,340 W	0,378 W	0,513 W
Γ	0,289	0,336	0,374
VSWR (perhitungan)	1,812.	2,012	2,195
VSWR (pengukuran)	2,7	4	1,7
Error VSWR	32,8 %	49,7 %	22,55 %